

Bully

Sisteme Distribuite

Laborator 3

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Fisierul de Log

Node 1 starts an election.

Node 1 sends election message to Node 2

Node 2 received election message from Node 1

Node 2 starts an election.

Node 2 sends election message to Node 3

Node 3 received election message from Node 2

Node 3 starts an election.

Node 3 is now the coordinator.

Node 1 acknowledges Node 3 as the new coordinator.

Node 2 acknowledges Node 3 as the new coordinator.

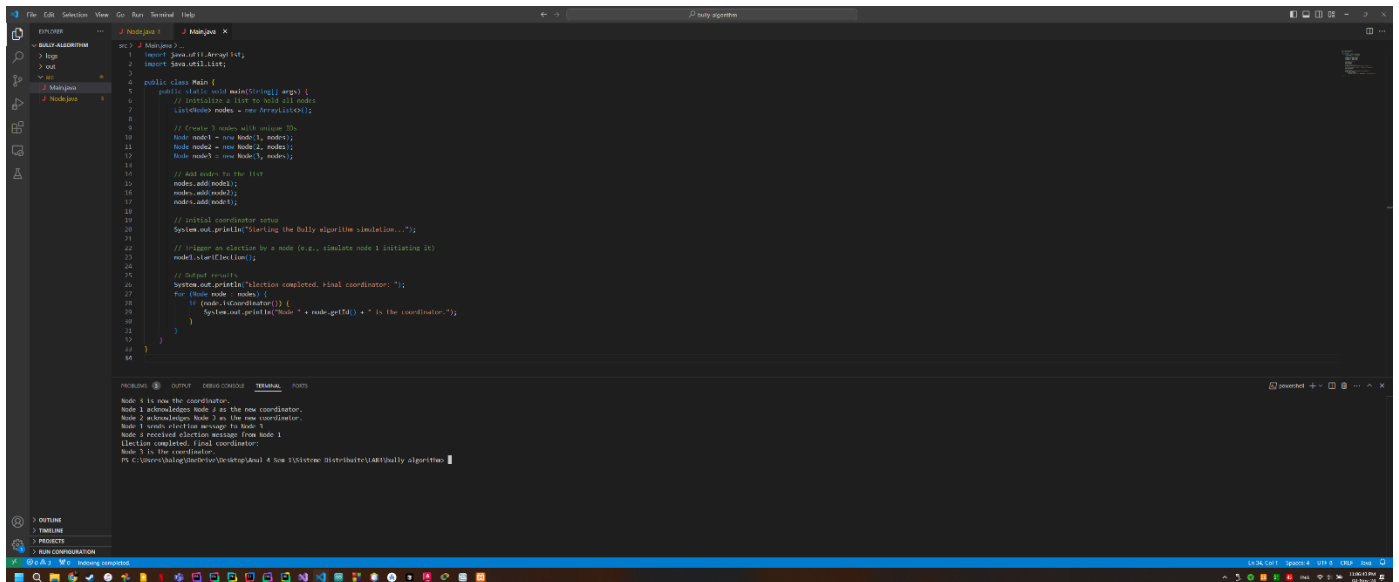
Node 1 sends election message to Node 3

Node 3 received election message from Node 1

Capturi ecran

```

1  import java.io.*;
2  import java.util.*;
3  import java.util.concurrent.*;
4
5  public class Node {
6      private int id;
7      private boolean lscCoordinator;
8      private List<Node> allNodes;
9
10     public Node(int id, List<Node> allNodes) {
11         this.id = id;
12         this.lscCoordinator = false;
13         this.allNodes = allNodes;
14     }
15
16     public int getId() {
17         return id;
18     }
19
20     public boolean lscCoordinator() {
21         return lscCoordinator;
22     }
23
24     public void setCoordinator(boolean lscCoordinator) {
25         this.lscCoordinator = lscCoordinator;
26         logEvent("Node " + id + " is now the coordinator.");
27     }
28
29     // Method to start an election
30     public void startElection() {
31         logEvent("Node " + id + " starts an election.");
32         boolean foundHigherId = false;
33
34         // Send election messages to nodes with higher ID
35         for (Node node : allNodes) {
36             // Send election message to node
37             // ... (code for sending message) ...
38         }
39
40         // Node 1 is now the coordinator.
41         // Node 2 acknowledges Node 1 as the new coordinator.
42         // Node 1 sends election message to Node 2.
43         // Node 2 receives election message from Node 1.
44         // Election completed. Node 1 is coordinator.
45         // Node 2 is the coordinator.
46         // PS: I have implemented a custom (non) 4-Node 1-Node 2-Node 3-Node 4-Node 5-Node 6-Node 7-Node 8-Node 9-Node 10-Node 11-Node 12-Node 13-Node 14-Node 15-Node 16-Node 17-Node 18-Node 19-Node 20-Node 21-Node 22-Node 23-Node 24-Node 25-Node 26-Node 27-Node 28-Node 29-Node 30-Node 31-Node 32-Node 33-Node 34-Node 35-Node 36-Node 37-Node 38-Node 39-Node 40-Node 41-Node 42-Node 43-Node 44-Node 45-Node 46-Node 47-Node 48-Node 49-Node 50-Node 51-Node 52-Node 53-Node 54-Node 55-Node 56-Node 57-Node 58-Node 59-Node 60-Node 61-Node 62-Node 63-Node 64-Node 65-Node 66-Node 67-Node 68-Node 69-Node 70-Node 71-Node 72-Node 73-Node 74-Node 75-Node 76-Node 77-Node 78-Node 79-Node 80-Node 81-Node 82-Node 83-Node 84-Node 85-Node 86-Node 87-Node 88-Node 89-Node 90-Node 91-Node 92-Node 93-Node 94-Node 95-Node 96-Node 97-Node 98-Node 99-Node 100-Node 101-Node 102-Node 103-Node 104-Node 105-Node 106-Node 107-Node 108-Node 109-Node 110-Node 111-Node 112-Node 113-Node 114-Node 115-Node 116-Node 117-Node 118-Node 119-Node 120-Node 121-Node 122-Node 123-Node 124-Node 125-Node 126-Node 127-Node 128-Node 129-Node 130-Node 131-Node 132-Node 133-Node 134-Node 135-Node 136-Node 137-Node 138-Node 139-Node 140-Node 141-Node 142-Node 143-Node 144-Node 145-Node 146-Node 147-Node 148-Node 149-Node 150-Node 151-Node 152-Node 153-Node 154-Node 155-Node 156-Node 157-Node 158-Node 159-Node 160-Node 161-Node 162-Node 163-Node 164-Node 165-Node 166-Node 167-Node 168-Node 169-Node 170-Node 171-Node 172-Node 173-Node 174-Node 175-Node 176-Node 177-Node 178-Node 179-Node 180-Node 181-Node 182-Node 183-Node 184-Node 185-Node 186-Node 187-Node 188-Node 189-Node 190-Node 191-Node 192-Node 193-Node 194-Node 195-Node 196-Node 197-Node 198-Node 199-Node 200-Node 201-Node 202-Node 203-Node 204-Node 205-Node 206-Node 207-Node 208-Node 209-Node 210-Node 211-Node 212-Node 213-Node 214-Node 215-Node 216-Node 217-Node 218-Node 219-Node 220-Node 221-Node 222-Node 223-Node 224-Node 225-Node 226-Node 227-Node 228-Node 229-Node 230-Node 231-Node 232-Node 233-Node 234-Node 235-Node 236-Node 237-Node 238-Node 239-Node 240-Node 241-Node 242-Node 243-Node 244-Node 245-Node 246-Node 247-Node 248-Node 249-Node 250-Node 251-Node 252-Node 253-Node 254-Node 255-Node 256-Node 257-Node 258-Node 259-Node 260-Node 261-Node 262-Node 263-Node 264-Node 265-Node 266-Node 267-Node 268-Node 269-Node 270-Node 271-Node 272-Node 273-Node 274-Node 275-Node 276-Node 277-Node 278-Node 279-Node 280-Node 281-Node 282-Node 283-Node 284-Node 285-Node 286-Node 287-Node 288-Node 289-Node 290-Node 291-Node 292-Node 293-Node 294-Node 295-Node 296-Node 297-Node 298-Node 299-Node 300-Node 301-Node 302-Node 303-Node 304-Node 305-Node 306-Node 307-Node 308-Node 309-Node 310-Node 311-Node 312-Node 313-Node 314-Node 315-Node 316-Node 317-Node 318-Node 319-Node 320-Node 321-Node 322-Node 323-Node 324-Node 325-Node 326-Node 327-Node 328-Node 329-Node 330-Node 331-Node 332-Node 333-Node 334-Node 335-Node 336-Node 337-Node 338-Node 339-Node 340-Node 341-Node 342-Node 343-Node 344-Node 345-Node 346-Node 347-Node 348-Node 349-Node 350-Node 351-Node 352-Node 353-Node 354-Node 355-Node 356-Node 357-Node 358-Node 359-Node 360-Node 361-Node 362-Node 363-Node 364-Node 365-Node 366-Node 367-Node 368-Node 369-Node 370-Node 371-Node 372-Node 373-Node 374-Node 375-Node 376-Node 377-Node 378-Node 379-Node 380-Node 381-Node 382-Node 383-Node 384-Node 385-Node 386-Node 387-Node 388-Node 389-Node 390-Node 391-Node 392-Node 393-Node 394-Node 395-Node 396-Node 397-Node 398-Node 399-Node 400-Node 401-Node 402-Node 403-Node 404-Node 405-Node 406-Node 407-Node 408-Node 409-Node 410-Node 411-Node 412-Node 413-Node 414-Node 415-Node 416-Node 417-Node 418-Node 419-Node 420-Node 421-Node 422-Node 423-Node 424-Node 425-Node 426-Node 427-Node 428-Node 429-Node 430-Node 431-Node 432-Node 433-Node 434-Node 435-Node 436-Node 437-Node 438-Node 439-Node 440-Node 441-Node 442-Node 443-Node 444-Node 445-Node 446-Node 447-Node 448-Node 449-Node 450-Node 451-Node 452-Node 453-Node 454-Node 455-Node 456-Node 457-Node 458-Node 459-Node 460-Node 461-Node 462-Node 463-Node 464-Node 465-Node 466-Node 467-Node 468-Node 469-Node 470-Node 471-Node 472-Node 473-Node 474-Node 475-Node 476-Node 477-Node 478-Node 479-Node 480-Node 481-Node 482-Node 483-Node 484-Node 485-Node 486-Node 487-Node 488-Node 489-Node 490-Node 491-Node 492-Node 493-Node 494-Node 495-Node 496-Node 497-Node 498-Node 499-Node 500-Node 501-Node 502-Node 503-Node 504-Node 505-Node 506-Node 507-Node 508-Node 509-Node 510-Node 511-Node 512-Node 513-Node 514-Node 515-Node 516-Node 517-Node 518-Node 519-Node 520-Node 521-Node 522-Node 523-Node 524-Node 525-Node 526-Node 527-Node 528-Node 529-Node 530-Node 531-Node 532-Node 533-Node 534-Node 535-Node 536-Node 537-Node 538-Node 539-Node 540-Node 541-Node 542-Node 543-Node 544-Node 545-Node 546-Node 547-Node 548-Node 549-Node 550-Node 551-Node 552-Node 553-Node 554-Node 555-Node 556-Node 557-Node 558-Node 559-Node 560-Node 561-Node 562-Node 563-Node 564-Node 565-Node 566-Node 567-Node 568-Node 569-Node 570-Node 571-Node 572-Node 573-Node 574-Node 575-Node 576-Node 577-Node 578-Node 579-Node 580-Node 581-Node 582-Node 583-Node 584-Node 585-Node 586-Node 587-Node 588-Node 589-Node 590-Node 591-Node 592-Node 593-Node 594-Node 595-Node 596-Node 597-Node 598-Node 599-Node 600-Node 601-Node 602-Node 603-Node 604-Node 605-Node 606-Node 607-Node 608-Node 609-Node 610-Node 611-Node 612-Node 613
```



Codul sursă

1. Main.java

```
import java.util.ArrayList;
```

```
import java.util.List;
```

```
public class Main {
```

```
    public static void main(String[] args) {
```

```
        // Initialize a list to hold all nodes
```

```
        List<Node> nodes = new ArrayList<>();
```

```
        // Create 3 nodes with unique IDs
```

```
        Node node1 = new Node(1, nodes);
```

```
        Node node2 = new Node(2, nodes);
```

```
        Node node3 = new Node(3, nodes);
```

```

// Add nodes to the list

nodes.add(node1);
nodes.add(node2);
nodes.add(node3);

// Initial coordinator setup
System.out.println("Starting the Bully algorithm simulation...");

// Trigger an election by a node (e.g., simulate node 1 initiating it)
node1.startElection();

// Output results
System.out.println("Election completed. Final coordinator: ");
for (Node node : nodes) {
    if (node.isCoordinator()) {
        System.out.println("Node " + node.getId() + " is the coordinator.");
    }
}
}
}

```

2. Node.java

```

import java.io.FileWriter;
import java.io.IOException;
import java.util.List;

public class Node {

    private int id;

    private boolean isCoordinator;

    private List<Node> allNodes;

```

```

public Node(int id, List<Node> allNodes) {
    this.id = id;
    this.isCoordinator = false;
    this.allNodes = allNodes;
}

public int getId() {
    return id;
}

public boolean isCoordinator() {
    return isCoordinator;
}

public void setCoordinator(boolean isCoordinator) {
    this.isCoordinator = isCoordinator;
    logEvent("Node " + id + " is now the coordinator.");
}

// Method to start an election
public void startElection() {
    logEvent("Node " + id + " starts an election.");
    boolean foundHigherId = false;
    // Send election messages to nodes with higher IDs
    for (Node node : allNodes) {
        if (node.getId() > this.id) {
            foundHigherId = true;
            logEvent("Node " + id + " sends election message to Node " + node.getId());
            node.receiveElectionMessage(this);
        }
    }

    // If no higher ID nodes are found, this node becomes the coordinator
    if (!foundHigherId) {

```

```

        setCoordinator(true);
        announceCoordinator();
    }
}

// Method to receive an election message
public void receiveElectionMessage(Node sender) {
    logEvent("Node " + id + " received election message from Node " + sender.getId());
    // Respond by starting its own election if not already a coordinator
    if (!isCoordinator) {
        startElection();
    }
}

// Announce this node as the new coordinator
private void announceCoordinator() {
    for (Node node : allNodes) {
        if (node.getId() != this.id) {
            node.receiveCoordinatorMessage(this);
        }
    }
}

// Receive coordinator announcement
public void receiveCoordinatorMessage(Node coordinator) {
    this.isCoordinator = false; // Reset coordinator status if receiving a message
    logEvent("Node " + id + " acknowledges Node " + coordinator.getId() + " as the new coordinator.");
}

// Log event method
private void logEvent(String message) {

```

```
System.out.println(message); // Print to console

try (FileWriter fw = new FileWriter("logs/event.log", true)) {

    fw.write(message + "\n");

} catch (IOException e) {

    e.printStackTrace();

}

}
```

Deficiențele observate

1. Gestionarea Erorilor de Comunicare: Dacă un nod este indisponibil sau nu răspunde, algoritmul nu are un mecanism rapid de detectare a acestui fapt, ceea ce poate duce la întârzieri în alegere.
2. Logare Limitată: Logurile sunt simple și nu conțin timestamp-uri sau informații detaliate care ar putea ajuta la urmărirea exactă a fiecărui pas al algoritmului și la diagnosticarea problemelor.

Soluții propuse

1. Introducerea Timeout-urilor: Adăugarea unui timeout la răspunsurile dintre noduri ar permite nodurilor să inițieze o nouă alegere dacă nu primesc un răspuns în timp util, prevenind blocajele.
2. Îmbunătățirea Logării: Adăugarea timestamp-urilor și a detaliilor specifice despre fiecare eveniment (de exemplu, tipul mesajului, nodul expeditor și receptor) ar permite o analiză mai detaliată și ar facilita identificarea problemelor de comunicare între noduri.